

L'Hôpital's Rule

Numberbender | WORKSHEET



Name: _____ Date: _____ Score: _____

Apply L'Hôpital's Rule where applicable. Show all work.

CALCULUS I Worksheet #47

L'Hôpital's Rule:	Suppose that $f(a)=g(a)=0$, that f & g are differentiable on an open interval containing a , and that $g'(x) \neq 0$ on ; then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$
Examples:	$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \frac{0}{0} \Rightarrow$ Use L'Hopital's Rule - Take derivative of numerator and denominator $\Rightarrow$ $\lim_{x \rightarrow 2} \frac{2x}{1} = \boxed{4}$
	$\lim_{x \rightarrow 1} \frac{x-1}{\sqrt{x}-1} = \frac{0}{0} \Rightarrow$ Use L'Hopitals: $\lim_{x \rightarrow 1} \frac{1}{\frac{1}{2\sqrt{x}}} = \lim_{x \rightarrow 1} \frac{2\sqrt{x}}{1} = \boxed{2}$
	$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x + 2} = \frac{0}{4} = \boxed{0} \Rightarrow$ You cannot use L'Hopital's Rule on this problem! Why??

Homework:

1	$\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 4}$	2	$\lim_{x \rightarrow 0} \frac{\sin 5x}{x}$
3	$\lim_{x \rightarrow 2} \frac{\sqrt{2+x} - 2}{x - 2}$	4	$\lim_{x \rightarrow 1} \frac{\sqrt[3]{x} - 1}{x - 1}$
5	$\lim_{x \rightarrow 2} \frac{x^2 - 4x + 4}{x^3 - 12x + 16}$	6	$\lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - \sin x}{1 + \cos 2x}$
7	$\lim_{x \rightarrow 1} \frac{x^3 - 1}{4x^3 - x - 3}$	8	$\lim_{x \rightarrow 3} \frac{x - 4}{x - 2}$
For 9-12: $\int_{-1}^0 f(x)dx = -2$ $\int_0^1 f(x)dx = 1$ $\int_{-1}^0 g(x)dx = 3$ $\int_{-1}^1 g(x)dx = 7$ $\int_{-1}^1 h(x)dx = 5$ $\int_0^1 h(x)dx = 11$			
9	$\int_{-1}^1 f(x)dx$	10	$\int_{-1}^0 h(x)dx$
11	$\int_{-1}^{-1} g(x)dx$	12	$3 \int_{-1}^0 g(x)dx - 2 \int_{-1}^0 h(x)dx$
13	If $f(x) = x^2 \tan^{-1} x$, then $f'(1) = ??$	14	Find the area between the curve $y^2 = 4x$ and the line $y=x$.
15	Find the area of the region enclosed by $y = e^x$, $y = \frac{1}{x}$, $x = 1$ and $x = 2$.	16	$\int_0^1 x(5^{-x^2})dx$
17	$\int_1^2 2^x dx$	18	$\sum_{j=-3}^3 (2j+1) =$

Answers:



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Answer key — for instructor use only.

CALCULUS I
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1) 3	2) 5	3) $\frac{1}{4}$	4) $\frac{1}{3}$
5) $\frac{1}{6}$	6) $\frac{1}{4}$	7) $\frac{3}{11}$	8) -1
9) -1	10) -6	11) 0	12) 21
13) $\frac{\pi+1}{2}$	14) $\frac{8}{3}$	15) $e^2 - e - \ln 2$	16) $\frac{4}{10\ln 5}$
17) $\frac{2}{\ln 2}$	18) 7		

